KNOWLEDGE ENGINEERING AND DATA SCIENCE

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Course Name	Knowledge Engineering and Data Science

EXPERIMENT 8: ONTOLOGY DEVELOPMENT USING PROTEGE

Aim

To create a simple ontology in Protégé by defining classes, subclasses, properties, and individuals.

Software Required

Protege Desktop.

Algorithm

1. Open Protege and create a new ontology.
2. Create classes such as Person, Student, and Teacher.
3. Make Student and Teacher subclasses of Person.
4. Create an object property such as studies With and a data property such as has Name.
5. Add individuals and assign suitable property values.
6. Save the ontology and check its structure.

Pseudocode

```
START
  Create a new ontology in Protégé
  Add classes and subclasses
  Add object and data properties
  Create individuals and assign property values
  Save ontology
END
```

Sample Input

Classes: Person, Student, Teacher; Individual: student1; Name: Meghana.

Expected Output

```
= RESTART: C:/Users/HP/AppData/Local/Programs/Python/Python313/ke exp 7 (1).py =
ONTOLOGY DEVELOPMENT
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Classes and Subclasses:
Person -> []
Student -> ['Person']
Teacher -> ['Person']

Individuals:
Meghana is a Student
Ravi is a Teacher

Relationships:
Meghana studies_with Ravi

Ontology created successfully!
```

Result

A basic university ontology is created successfully using Protege.

EXPERIMENT 9: KNOWLEDGE GRAPH CONSTRUCTION USING PYTHON

Aim

To construct and display a simple knowledge graph using Python and the Network X library.

Software Required

Python 3, NetworkX, and Matplotlib.

Algorithm

7. Import NetworkX and Matplotlib.
8. Create an empty graph.
9. Add entities as nodes.
10. Add relationships as edges.
11. Draw the graph with labels.
12. Display the graph.

Pseudocode

```
START
  Import NetworkX and Matplotlib
  Create a directed graph
  Add entity nodes and relationship edges
  Arrange nodes and draw graph with labels
  Display graph
END
```

Sample Input

Entities: Meghana, Python, Programming. Relationships: learns, is a type of.

Expected Output

```
=== RESTART: C:/Users/HP/AppData/Local/Programs/Python/Python313/ke exp 8.py ===
KNOWLEDGE GRAPH
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Meghana -- learns --> Python
Python -- is a type of --> Programming

Knowledge graph created successfully!
```

Result

A simple knowledge graph is constructed and visualized using Python.